DO NOW — RECALL

Operating-system functions

Name: _____ Class: _____ Date: _____

1. What does LMC LDA do?

2. What does the CPU control unit do?

3. Define a process.

4. Binary search reduces a sorted 32-element candidate set to one. How many successive halvings is this?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Loads ACC from a mailbox.

2. Model answer

Coordinates and decodes execution.

3. Model answer

An instance of a program being executed or scheduled.

4. Model answer

5: 32→16→8→4→2→1.

Vocabulary — know these terms

process — Running/scheduled program instance

filesystem — Organised file storage

driver — Device interface software

spooling — Queued peripheral work

NEW LEARNING

Read, annotate and explain

Understand

The OS mediates application requests and access to hardware.

Trace

Resource management schedules processes, allocates memory, controls files and drivers.

Explain

Permissions and isolation help keep processes and users from accessing unauthorised data.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A pupil app requests a protected file. Explain the OS steps.

Predict first

The OS checks authorisation before returning content; a missing permission should prevent the read.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State four operating-system responsibilities. [4]

Question 2

Describe a protected file read from application request to result. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain how an OS manages a browser, game and backup concurrently, including protection. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Applications must directly control all hardware to read files.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.